A.I., Machine Learning, and Ethical Decision-Making in Sustainable Infrastructure: A Hurricane Sandy Modeling Perspective

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Abstract

Geohazards become disasters when physical hazard events intersect with vulnerable populations, infrastructure, and societal systems. As data-driven approaches increasingly inform disaster assessment and infrastructure planning, the prioritization of disaster severity becomes dependent on analytical design choices rather than purely objective measurement. This study uses the EM-DAT global disaster database to examine how modeling frameworks and variable selection shape disaster prioritization outcomes.

Using R, we compare a multiple linear regression model with a Random Forest model to evaluate how conventional impact indicators including total deaths, total affected population, and economic loss, alongside contextual variables, contribute to a composite disaster severity score (Cho, M. et al., 2026). An ethical stress-test is applied by comparing full models with reduced models that exclude dominant impact indicators. Results show that when conventional metrics are included, both models strongly prioritize measurable impacts. When these indicators are removed, importance redistributes toward duration, rare high-consequence events, data completeness, and regional context. Random Forest results show strong concentration in conventional impact variables, while reduced models elevate rare high-consequence events and duration as dominant predictors.

Hurricane Sandy (2012) is used as a regional case to illustrate how classification structure and variable inclusion influence algorithmic visibility. While Sandy ranks consistently within the upper percentiles across all frameworks, its relative position varies depending on model structure, highlighting how methodological choices shape disaster prioritization. These findings emphasize that AI-informed assessments reflect structured analytical perspectives.

Introduction

Natural hazards such as floods, earthquakes, storms, and droughts vary in physical magnitude and spatial extent, yet their societal impacts depend heavily on exposure, infrastructure resilience, and the broader systems through which geohazard impacts unfold. As algorithmic and data-driven approaches become more influential in disaster assessment, funding allocation, and infrastructure planning, the prioritization of different indicators of disaster severity becomes complex and method-dependent. Common disaster metrics such as economic loss, mortality, and affected population are useful because they are measurable and comparable across events. At the same time, these metrics embed assumptions about what counts most, which means model-based assessments can reproduce value choices even when they appear technically objective.

Across the New York City and Long Island region, this question is especially relevant because coastal hazards are often experienced as compound events involving storm surge, flooding, and shoreline morphology. Hurricane Sandy remains one of the most consequential coastal disasters affecting transportation systems, electrical infrastructure, hospitals, housing, and communities across the area. Therefore, examining how different analytical approaches structure disaster prioritization provides meaningful insight into how measurable indicators influence model-based assessments of disaster severity and how methodological choices shape what becomes legible as "severe" within algorithmic systems.

Recent advances in artificial intelligence and geospatial analytics have expanded the application of machine learning techniques in natural hazard assessment and forecasting. Previous research has demonstrated the effectiveness of artificial neural networks and hybrid statistical–machine learning approaches for flood prediction and hazard monitoring (Tsakiri et al., 2018; Marsellos et al., 2018; Mahmood et al., 2026). These developments highlight the growing role of AI-driven modeling frameworks in supporting decision-making for sustainable infrastructure and risk mitigation, particularly in coastal regions where compound hazards interact with geological processes.

Hurricane Sandy and Coastal New York

Coastal geohazards in the New York and Long Island region are strongly influenced by geomorphology, barrier island systems, shallow continental shelf bathymetry, and storm surge amplification along the coastline. Storm-driven coastal flooding is therefore not solely a meteorological phenomenon as it also interacts with geological and geomorphic processes that shape hazard propagation and infrastructure vulnerability.

As a result, Hurricane Sandy (2012) provides a critical case for examining how algorithmic prioritization may interact with disaster taxonomy. Sandy produces extensive coastal flooding, infrastructure failure, power outages, transportation disruption, and long-term economic impacts across the region. With global disaster databases such as EM-DAT, Sandy is primarily classified as a tropical cyclone. That classification is meteorologically appropriate, but it does not fully capture the geological amplification processes through which storm surge translated into severe coastal flooding across the New York Bight.

This distinction matters because analysts often filter or subset disaster records using database categories. If coastal flood analyses rely on records labeled as floods, hurricane-driven surge events such as Sandy may be underrepresented within the training data used for algorithmic prioritization. Moreover, database categories may not fully represent the compound coastal processes that determine regional hazard severity and the classification systems that inform geohazard decision-making. Using Sandy as a regional study helps connect the broader global modeling exercise in evaluating how classification structure and variable inclusion may affect the visibility of regionally significant coastal hazards in AI-informed data interpretation.

Methods

Data

This study uses the EM-DAT International Disaster Database maintained by the Centre for Research on the Epidemiology of Disasters (CRED). The working dataset contains 10,640 global disaster records obtained through a custom EM-DAT query generated on February 12, 2026. Variables include disaster classification, geographic location, temporal attributes, hazard descriptors, reported human impacts, economic losses, and contextual economic indicators.

Data preprocessing was conducted in R (RStudio). Variable names were standardized, relevant fields were selected, and temporal variables were used to reconstruct approximate event start and end dates. Where month or day values were missing, default values were inserted to permit date construction. Event duration was then calculated as the difference between reconstructed end and start dates measured in days. For economic loss variables, inflation-adjusted EM-DAT values were prioritized where available.

Several derived variables were created for modeling: a composite severity score based on z-standardized log-transformed deaths, total affected population, and economic damage; duration in days; data completeness; damage per affected person; and a binary rare high-consequence indicator defined as events above the 95th percentile of either total deaths or total economic damage. Sparse categorical levels were consolidated, numeric missing values were median-imputed, and missing categorical values were retained as an “Unknown” class. Multi-variable environmental modeling frameworks similarly emphasize that integrating diverse risk indicators inherently involves trade-offs, where the relative influence or “weight” of each variable is shaped by model design choices (Ai et al., 2026). This reinforces the importance of examining how different weighting structures influence final outputs.

Analytical Workflow

The analysis proceeded in four stages. First, the cleaned EM-DAT dataset was inspected to ensure variables were appropriately formatted for modeling. Second, a multiple linear regression model was fit using the composite severity score as the dependent variable; absolute t statistics were used as a practical indicator of predictor influence. Third, a Random Forest model was fitted using the same dependent variable, and a predictor of importance was extracted using percent increase in mean squared error (%IncMSE).

Fourth, an ethical stress-test design was applied by estimating both full and reduced versions of the linear model and the Random Forest model. The full models included dominant conventional impact metrics such as total deaths, total affected population, and total economic damage. The reduced models removed those variables to evaluate how model behavior changed when conventional measurable indicators were withheld. This comparison was designed to test whether the models would elevate alternative contextual variables when dominant metrics were absent.

Machine learning approaches are increasingly used in disaster management due to their ability to process large, multimodel datasets and support tasks such as classification, prediction, and post-disaster analysis (Chamola, et al., 2021). These models are well-suited for identifying patterns across variables such as population impact, economic loss, and hazard characteristics, though their outputs remain highly dependent on input structure and feature selection. The methodological framework aligns with established applications of machine learning and neural network models in geohazard forecasting and environmental monitoring (Tsakiri et al., 2018; Marsellos et al., 2018). Such approaches have demonstrated strong capability in identifying nonlinear relationships and temporal patterns within geospatial datasets, supporting both predictive modeling and decision-support applications in hazard-prone regions.

Ensemble Prioritization Framework

To evaluate whether integration of optimization-based and ethical prioritization produces more balanced disaster ranking outcomes, the study proposes an ensemble severity framework in which an AI-derived priority score is combined with a human ethical severity score using a weighted linear combination:

$$\textit{Ensemble Score} = \alpha(\textit{AI Priority Score}) + (1 - \alpha)(\textit{Human Ethical Score})$$

Where α represents the relative weight assigned to algorithmic optimization. In this study, equal weighting ($\alpha = 0.5$) was applied to reflect a balanced integration of measurable magnitude and ethical sensitivity. By comparing rankings across AI-only, ethical-only, and ensemble frameworks, this approach evaluates whether hybrid decision-support models may provide more robust guidance for disaster governance and sustainable coastal infrastructure planning.

Prior research has demonstrated that hybrid decision-making systems, where human judgment and algorithmic outputs are combined, can outperform in complex prioritization tasks. This study's ensemble weighting framework aligns with Cruz (2024) conceptually in treating model outputs not as definitive answers but as components of a broader evaluative structure. Disaster assessment further demonstrates that AI-classification systems are often used to directly inform real-time prioritization decisions in identifying areas of severe damage from social media imagery (Zhang et al., 2020). However, purely automated systems can misidentify critical regions or misclassify severity levels leading to misallocation of response resources. As a result, hybrid human-AI systems have been proposed to improve accuracy by incorporating human judgment into model outputs, reinforcing the importance of interpretability and oversight in disaster-related decision-making.

Results

Model priorities shift substantially when dominant impact indicators are removed. That shift is strongest in the Random Forest results, which show how the definition of severity depends on the information structure made available to the model. To evaluate how variable inclusion influences model behavior, Random Forest variable importance was compared between full and reduced models (Figure 1).

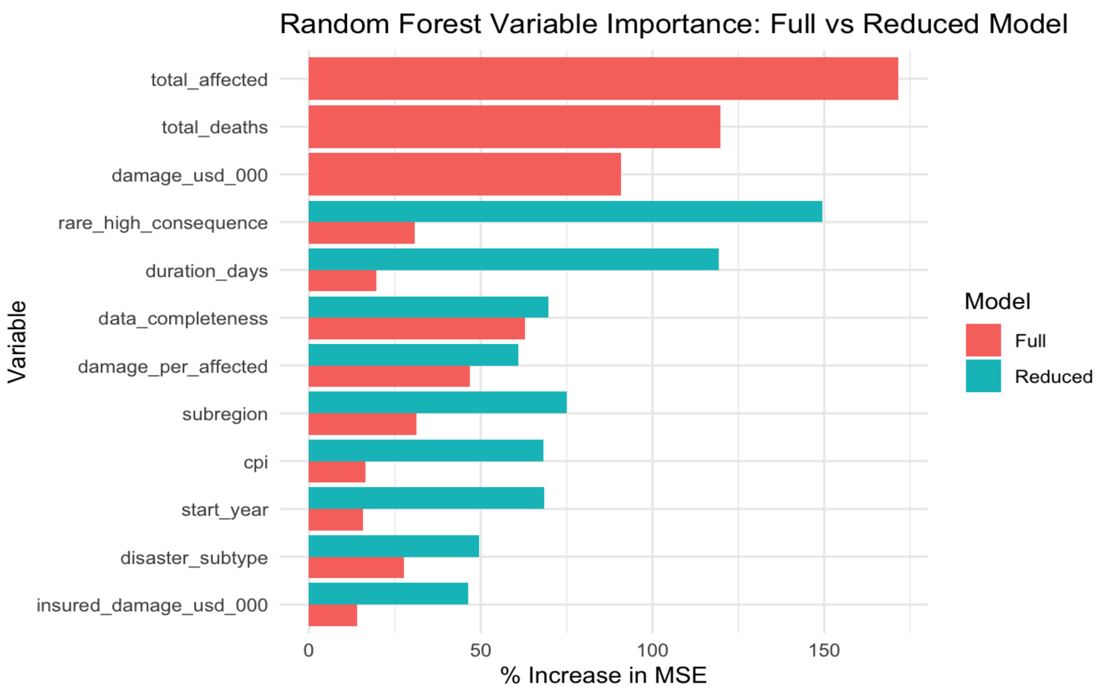

***Figure 1**. Random Forest variable importance comparison between full and reduced models.*

In the full Random Forest model, the dominant predictors were total affected population (%IncMSE = 171.57), total deaths (%IncMSE = 119.73), and economic damage (%IncMSE = 90.74). Secondary contributors included data completeness (%IncMSE = 62.98), damage per affected (%IncMSE = 46.78), and subregion (%IncMSE = 31.24). These outputs indicate that the model concentrates strongly around conventional measurable impact metrics when those variables are present. The shift in predictor importance between full and reduced models is summarized in Table 1.

***Table 1**. Top Random Forest predictors in full and reduced models based on %IncMSE.*

Full model term	%IncMSE	Reduced model term	%IncMSE

total_affected	171.57	rare_high_consequence	149.42
total_deaths	119.73	duration_days	119.40
damage_usd_000	90.74	subregion	74.92
data_completeness	62.98	data_completeness	69.67
damage_per_affected	46.78	start_year	68.56
subregion	31.24	cpi	68.34

Table 1 quantifies the shift in predictor importance, highlighting the transition from conventional impact metrics to contextual and proxy variables. In the reduced Random Forest model, importance redistributed. Rare high-consequence status became the strongest predictor (%IncMSE = 149.42), followed by duration in days (%IncMSE = 119.40), subregion (%IncMSE = 74.92), data completeness (%IncMSE = 69.67), start year (%IncMSE = 68.56), and CPI (%IncMSE = 68.34). This pattern demonstrates when dominant severity indicators are removed, the model does not become neutral but instead reorients toward alternative contextual and proxy variables.

The linear model displayed a similar but not identical pattern. To further examine differences in model behavior, variable importance from the linear model and Random Forest model were compared directly (Figure 2). In the full linear model, strong contributors included the rare high-consequence indicator ($|t| = 38.27$), disaster type = Wildfire ($|t| = 13.26$), duration in days ($|t| = 13.01$), region = Europe ($|t| = 12.03$), total affected population ($|t| = 10.32$), and total deaths ($|t| = 8.07$). In the reduced linear model, the rare high-consequence indicator became even stronger (|t|

= 41.76), while duration remained prominent (|t| = 13.19). Variable importance from the linear and Random Forest models were compared directly (Figure 2).

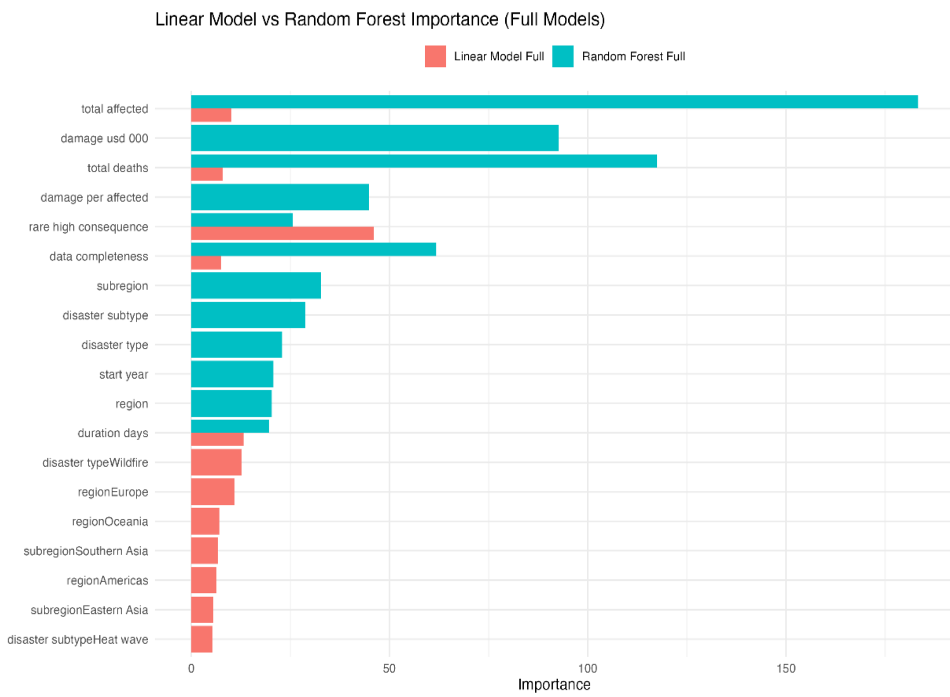

***Figure 2**. Comparison of variable importance between linear regression and Random Forest models (full model).*

The Random Forest model concentrates importance within measurable impact variables, while the linear model distributes influence more broadly across categorical and contextual predictors.

One useful contrast between frameworks appears in reconstruction cost. In the full linear model, reconstruction_usd_000 had a modest positive importance value (|t| = 1.17), whereas in the full Random Forest model it showed a negative %IncMSE value (-0.74). This suggests that a variable can appear contributory under linear assumptions while adding little or even slightly harming predictive performance in a nonlinear interaction-based model. Model performance was evaluated using observed versus predicted comparisons to assess predictive consistency across modeling frameworks (Figure 3).

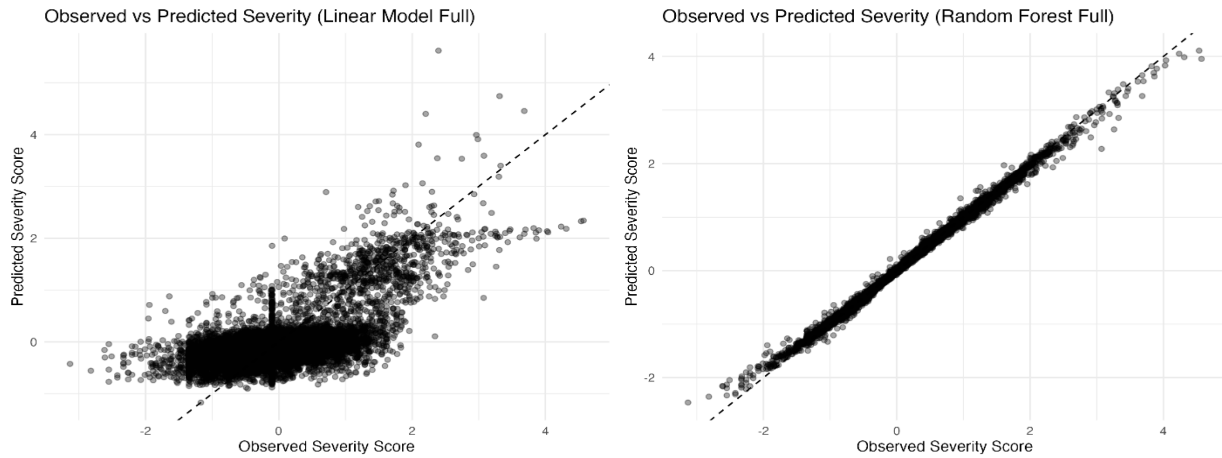

***Figure 3**. Observed versus predicted severity scores for linear and Random Forest models.*

The Random Forest model demonstrates tighter clustering around the 1:1 relationship, indicating stronger predictive performance relative to the linear model. This suggests improved predictive consistency, but does not necessarily imply that the model captures a more comprehensive or unbiased representation of disaster severity. These differences highlight how model selection can substantially influence the representation of disaster severity, with implications for prioritization and decision-making frameworks.

Hurricane Sandy Case Study

To assess how model structure shapes regional prioritization, Hurricane Sandy (2012) was evaluated across AI-only, ethical-only, and ensemble ranking frameworks. Table 2 summarizes Sandy's severity score, global rank, and percentile position under each framework.

This comparison provides a focused regional test of the broader modeling argument by illustrating how the same disaster shifts in relative prominence depending on how severity is defined analytically.

***Table 2**. Hurricane Sandy ranking across modeling frameworks.*

Framework	Severity Score	Global Rank	Percentile (%)
RF Full	1.863	528	95.05
LM Full	3.480	193	98.2
Ethical Only	1.026	523	95.09
Ensemble RF	1.445	472	95.57
Ensemble LM	2.253	227	97.88

Across all frameworks, Hurricane Sandy ranks within the upper percentiles of global disaster severity. However, variation in rank position across frameworks demonstrates that prioritization is sensitive to model structure and variable inclusion. The higher ranking under the full and ensemble models suggests that additive model structures and blended ethical weighting may amplify Sandy's importance compared to Random Forest-based rankings. Although Sandy remains consistently visible as a high-severity event, these differences indicate that its prominence is not fixed but contingent on how severity is defined within each framework. This reinforces that regionally significant coastal disasters may be consistently recognized as severe, while still being differentially emphasized depending on modeling assumptions and prioritization logic.

Beyond its percentile ranking, Sandy's impact profile reflects the interaction between meteorological forcasting and coastal geological configuration. The shallow continental shelf, concave New York Bight geometry, barrier island systems, and low-lying sedimentary coastal plains contributed to surge amplification and prolonged inundation. While these geological factors are not directly encoded as predictors within the EM-DAT database, they materially influenced the disaster's severity metrics. This highlights a structural limitation of database-driven modeling: geological amplification processes may shape outcomes without being explicitly represented within algorithmic inputs.

Discussion

The core interpretive argument is that the models are not neutral. They do not simply detect disaster severity, but rather operationalize severity according to variables, proxies, and classifications made available to them. In the full model, severity is dominated by conventional measurable impacts. This pattern is clearly illustrated in Figure 1 and Figure 2, where Random Forest importance is concentrated among impact variables, while linear model influence remains more distributed. Model performance differences observed in Figure 3 further reinforce that predictive accuracy and variable prioritization are not equivalent, as models may perform well while emphasizing different drivers of severity. In the reduced models, the same analytical systems shift toward duration, rarity, completeness, timing, and regional context. As shown in Figure 1, this redistribution highlights how model outputs are contingent on available inputs rather than reflecting an intrinsic definition of disaster severity (Schneider et al., 2025). This means that prioritization is structurally contingent rather than fixed. Notably, the Random Forest model concentrates importance within measurable impact variables, whereas the linear model distributes influence more broadly across categorical and contextual predictors, highlighting how different modeling frameworks operationalize severity in structurally distinct ways.

This matters because algorithmic systems may reinforce the visibility of disasters that are well documented, economically quantified, or produced through standardized metrics, while comparatively obscuring events in data-limited regions or those whose impacts are expressed through more indirect forms of vulnerability. The reliance of data completeness in Random Forest outputs hints that reporting structure itself may influence what becomes prioritized. These differences have ethical implications. Similar dynamics have been observed in healthcare contexts, where AI-driven triage systems have been shown to influence the allocation of limited resources in ways that can unintentionally reinforce bias or misprioritize need (Costa et al., 2025). These findings parallel the observed discrepancies within this study, suggesting that algorithmic prioritization can produce materially different outcomes depending on how variables are weighted and interpreted. Furthermore, Ghaffarian's (2023) study aligns with broader concerns in risk management regarding the "black box" nature of AI systems. While machine learning models can significantly enhance decision-making, their lack of transparency can obscure how and why certain outcomes are prioritized.

Hurricane Sandy helps make this issue tangible. Standard database categories do not necessarily capture how coastal impacts are amplified by shoreline form, barrier systems, and the geometry of the New York Bight. Within this perspective, a storm category can be meteorologically accurate while still under expressing the flood-producing geological context most relevant to local infrastructure planning. When evaluated together, the statistical comparison and the Sandy case support a broader methodological claim: disaster ranking outcomes are shaped by variable inclusion, data representation, and classification structure. Any use of AI or machine learning in sustainable infrastructure planning should therefore make those assumptions explicit rather than treating rankings as self-evident. The Sandy case further demonstrates that even highly consequential regional events are subject to shifts in prioritization depending on classification structure and variable inclusion, reinforcing the need for transparency in AI-informed decision frameworks.

The integration of artificial intelligence with geological understanding represents a critical advancement in geohazard assessment. Previous studies utilizing neural networks, geostatistical modeling, and remote sensing techniques have demonstrated the capacity of AI-driven systems to enhance hazard detection, monitoring, and forecasting in complex geological environments (Marsellos et al., 2018; Tsakiri et al., 2018). The present study extends this interdisciplinary approach by highlighting how algorithmic prioritization interacts with coastal geomorphology and disaster taxonomy, highlighting the importance of integrating geological context within AI-informed infrastructure planning frameworks.

Conclusion

This study shows that algorithmic disaster prioritization is sensitive to variable inclusion, modeling framework, and classification structure. Random Forest models strongly privilege total affected population, total deaths, and economic damage when those variables are available, but shift toward rare high-consequence, duration, and contextual indicators when conventional metrics are removed. These findings emphasize that AI-informed disaster assessment should be interpreted as a structured analytical perspective rather than an objective reflection of real-world severity.

Regionally, Hurricane Sandy provides a useful interpretive case because it demonstrates how coastal hazard severity can be shaped by compound interactions among meteorology, geomorphology, and infrastructure exposure. Meteorological classification is technically accurate; however, coastal geohazard severity emerges through compound interaction among atmospheric forcing, geomorphological structure, and infrastructure exposure. Algorithmic systems that rely strictly on categorical taxonomy may not fully capture this compound interaction.

Together, these findings demonstrate that disaster prioritization is not solely a function of hazard magnitude but rather a product of how data, classification systems, and modeling frameworks interact to define what is visible within analytical systems.

Credit Authorship Contribution Statement

Izaguirre Cortez, B.: RStudio coding, conceptualization, methodology, model development, visualization, and paper composition. Rezki, A.: review, editing, and references. Najarro Lima, M.: review and editing. Marsellos, A. E.: supervision, conceptual guidance, contribution to the Hurricane Sandy case study and Discussion, regional context, references, review and editing.

Software and Packages

All analyses were conducted using R version R-4.5.2 statistical software within the RStudio environment. The following R packages were utilized: readxl for importing excel files, dply, tidyr, tibble, and purrr for data manipulation, janitor for variable name standardization, lubridate for date construction and duration calculations, forcats for categorical variable handling, randomForest for machine learning modeling, broom for regression output formatting, ggplot2 for visualization, vip for variable importance support. Package citations were generated using the citation("package_name") function in R and are included in the References section.

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